

Stovetop Monitor

Build and Install Manual

By: Jim Schrempp and Bob Glicksman; v1.1, 8/13/2026

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https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Documents/Terms_of_Use_License_and_Disclaimer.pdf

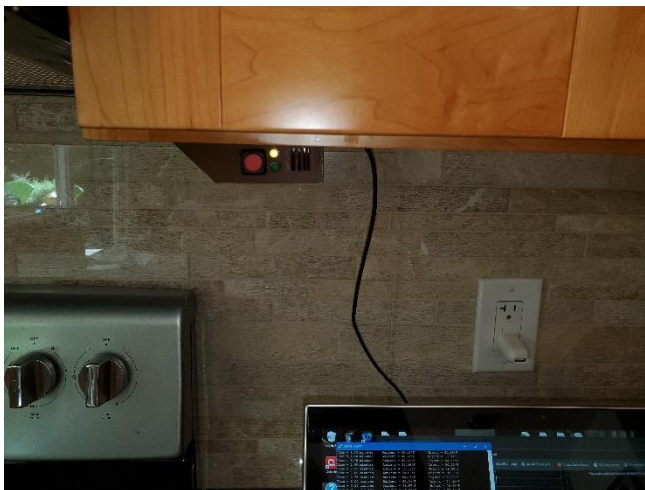


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DOCUMENT OVERVIEW.

This document describes the steps necessary to build and install the Stovetop Monitor from Team Practical Projects. The Stovetop Monitor monitors the state of a stovetop and alarms if it senses that a burner is turned on for too long.

The Stovetop Monitor consists of the following components:

- Hardware. The electronics and 3D printed enclosure that need to be mounted looking down at the stovetop being monitored.
- Firmware. The embedded software in the Particle Photon 2 microcontroller that is part of the electronic hardware.
- Mobile Phone Alarm. A Particle Webhook and the Pushover App are used to generate an alarm notification on the user's mobile phone.
- Mobile Phone App. An App that can be installed on the user's mobile phone allowing the user to monitor the state of the Stovetop Monitor remotely, and to reset an alarm that is in process.

See the "Overview and Theory of Operation Manual" for details about the system architecture and algorithm of the Stovetop Monitor project:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Stovetop_Monitor_Overview_and_Theory_of_Operation.pdf

Also see the "User Manual" for instructions describing local and remote control and status of the Stovetop Monitor:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Stovetop_Monitor_Overview_and_Theory_of_Operation.pdf

STOVETOP MONITOR HARDWARE.

The Stovetop Monitor hardware consists of a custom printed circuit board (PCB) and a custom 3D printed enclosure. The PCB holds the electronics, including the Particle Photon 2 microcontroller. The 3D printed enclosure has 2 parts – a Top cover that is for mounting the Stovetop Sensor underneath a cabinet, and the Enclosure body that holds all of the electronics, controls and indicators and the IR temperature sensor module.

STOVETOP MONITOR ELECTRONICS.

Most of the electronic components are mounted on a custom PCB. This PCB was originally developed for another Team Practical Projects project; see:

https://github.com/TeamPracticalProjects/MN_ACL/tree/master/Hardware/PC_board

Not all of the components on the original PCB design are required for this project. Figure 1, below, shows those components that are needed for the Stovetop Monitor and Figure 2 shows the components that should not be installed on the PCB for this project.

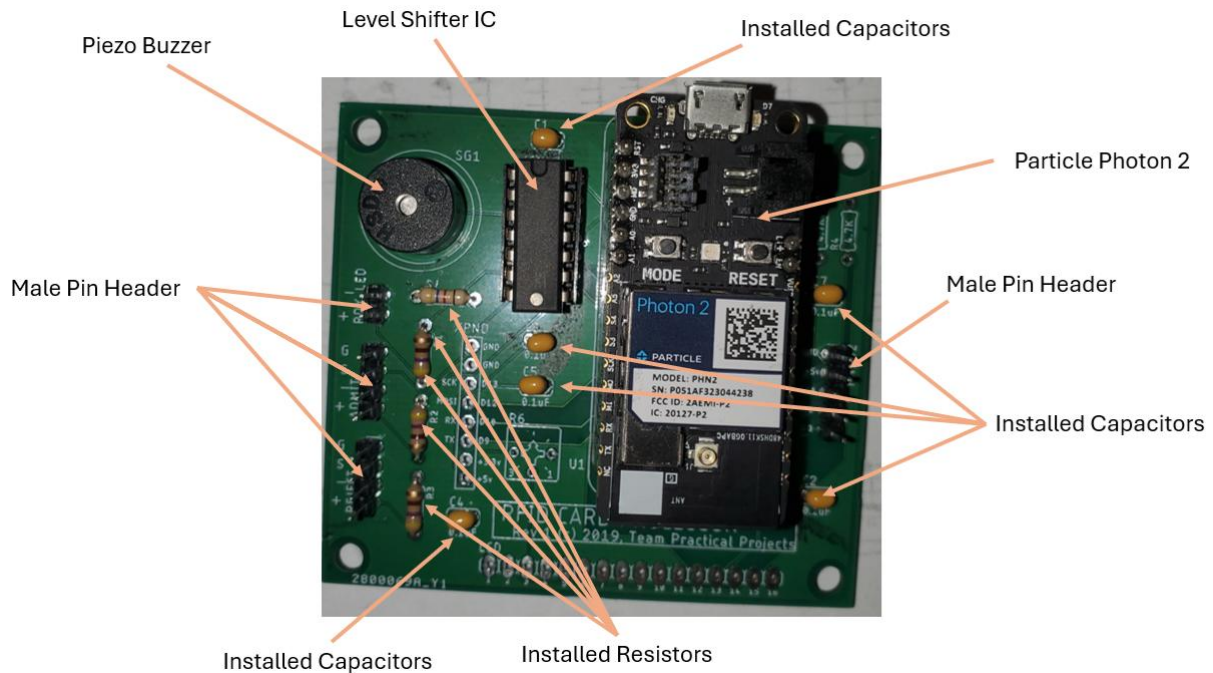


Figure 1. Electronics PCB – Installed Components.

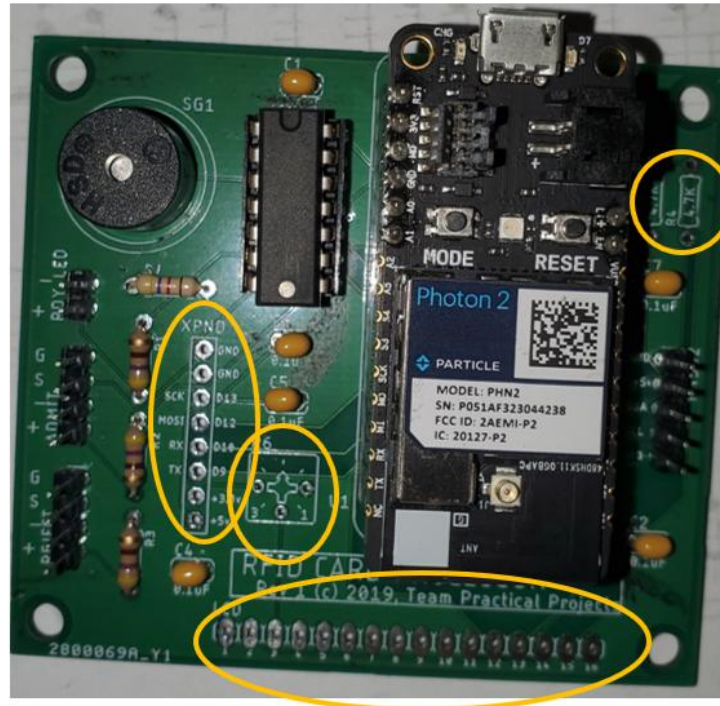


Figure 2. Components NOT Installed on PCB.

In particular, resistors R4 and R5 must not be installed on the PCB for this project, as the sensor module used already includes the I2C bus pull up resistors.

The PCB schematic is here:

https://github.com/TeamPracticalProjects/MN_ACL/blob/master/Hardware/PC_board/RFID_R1_Schematic.pdf

A parts list that includes the components for the PCB is here:

https://github.com/TeamPracticalProjects/MN_ACL/blob/master/Hardware/RFID_Station_Parts_List.pdf

NOTE: At the time of this writing, there is an issue with the Piezo Buzzer listed on this parts list (Adafruit P/N 1536). Adafruit is aware of the problem but it remains unresolved at this time. See the file “A Note About Piezo Buzzers.txt” for alternatives:

https://github.com/TeamPracticalProjects/MN_ACL/blob/master/Hardware/A%20NOTE%20ABOUT%20PIEZO%20BUZZERS.txt

In addition to the parts mounted on the PCB, the following additional electronic parts are to be mounted inside the enclosure and wired to the male pin header connectors on the PCB:

- Generic 5 mm green LED (plus 5 mm LED panel mounting hardware)
- Generic 5 mm yellow LED (plus 5 mm LED panel mounting hardware)
- Red Backlit Momentary Pushbutton Switch (<https://www.adafruit.com/product/1439>)
- IR Temperature Sensor Module:
(https://www.amazon.com/dp/B0B63K5V7T?ref=ppx_yo2ov_dt_b_fed_asin_title)

Note that the IR Temperature Sensor module comes with straight pin headers, but we have chosen to solder on 90 degree right angle pin headers to make the cabling a little neater.

STOVETOP MONITOR ENCLOSURE ASSEMBLY.

The Stovetop Monitor Enclosure consists of two 3D printed parts:

The Enclosure, shown in figure 3, below, houses all of the electronics. The 3D CAD file (.stl format) can be found in the “Hardware” folder of this repository:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Hardware/stove%20monitor%20box.stl

The Enclosure is open but has tabs that engage with the Top cover. The Top piece has a number of screw holes in the top surface and in the right angle side, allowing for flexible mounting underneath a cabinet that overlooks the stovetop. The 3D CAD file (.stl format) can be found in the Hardware folder of this repository:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Hardware/stove%20monitor%20mount.stl

These two files can be imported into any slicer program to produce G-code files for 3D printing. We printed these components at 0.2mm resolution in generic PLA, but PETG or ABS might be preferred if the Stovetop Monitor is located where it might be exposed to high heat.

All of the Stovetop Monitor components are assembled inside the Enclosure, as shown in figure 3. There are 4 standoffs that align with the 4 mounting holes on the printed circuit board that is described in the Stovetop Monitor Electronics section, above. M3 screws may be used to mount the printed circuit board securely into the Enclosure. The printed circuit board is positioned so that the micro USB connector on the Photon 2 microcontroller sticks through the slot in the rear of the Enclosure. A USB cable connected to the Photon 2's USB port supplies power to the electronics and also provides access to serial port data. Serial port data may be collected and analyzed to customize the thresholds and timeouts in the Photon 2 firmware to optimize the Stovetop Monitor performance to a specific installation and routine cooking conditions.

The Enclosure has a 45 degree sloping side that is designed to hold the IR temperature sensor module. There is a hole in this side where the sensor itself protrudes through the enclosure. The 45 degree angle exists so that the sensor looks down at the stovetop and the sensor's 90 degree field of view covers the entire stovetop. The IR temperature sensor module is held in place using ordinary electrical tape, as shown in figure 3.

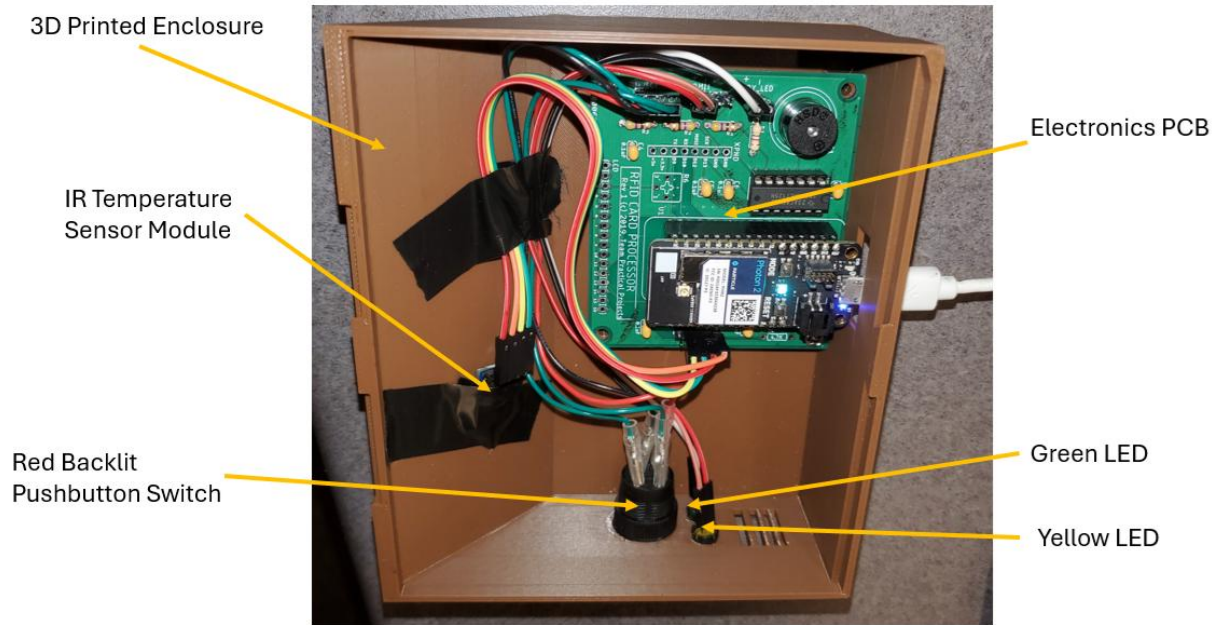


Figure 3. Assembled Enclosure.

The front side of the Enclosure has mounting holes for the three additional electronic components:

- Green LED
- Yellow LED
- Red backlit pushbutton switch

We recommend using standard 5 mm LED panel mounting hardware to mount the green and yellow LEDs to the enclosure. The red pushbutton switch mounts to the Enclosure using its own backset nut. Figure 4 is a front view of these components mounted in the Enclosure.

The front of the Enclosure also has grill slits that allow the buzzer sound to escape from the front of the Enclosure.

Standard female – female Dupont jumper cables can be used to connect the LEDs and the IR temperature sensor module pins to the male header pins on the printed circuit board, as shown in figure 3.

We made our own connector cables to connect the pushbutton switch's spade lugs to the printed circuit board header pins. We used #26 awg stranded wire for these cables. We used 2.8 mm spade connectors for the pushbutton switch end of the wires and female Dupont crimp connectors for the PCB end of these wires. The crimp spade lugs can be obtained in the following kit:

<https://www.adafruit.com/product/4748>

The crimp Dupont female connectors and plastic connector bodies can be obtained from this kit:

https://www.amazon.com/dp/B0DRK17PTW/ref=sspa_dk_rhf_yoy_pt_sub_2/?encoding=UTF8&ie=UTF8&sp_csd=d2lkZ2V0TmFtZT1zcF9yaGZfeW95&pd_rd_w=MD9fd&content-id=amzn1.sym.69f8f66e-98eb-45f4-92c0-0c9f99faf570&pf_rd_p=69f8f66e-98eb-45f4-92c0-0c9f99faf570&pf_rd_r=AVH4TYG53XQQ112G30G3&pd_rd_wg=vUs0M&pd_rd_r=b373cd7d-91aa-4d6e-96fe-3ec430079ca3&ref=sspa_dk_rhf_yoy_pt_sub&th=1

The crimping tool that comes with this kit can be used to crimp both the Dupont connector pins and the spade lugs.

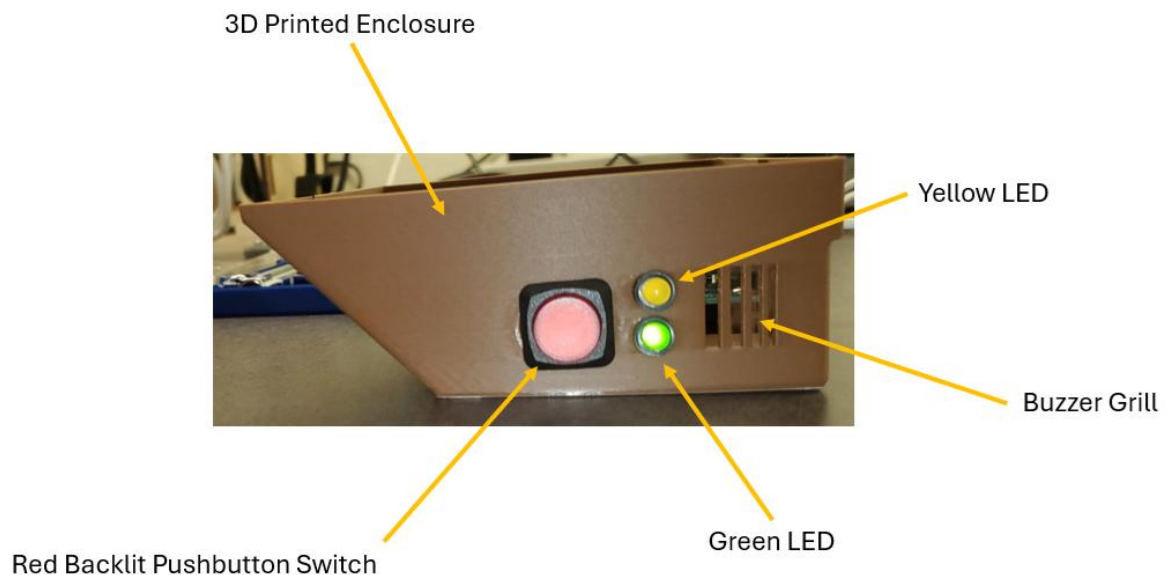


Figure 4. Assembled Enclosure Front Panel.

The parts that are mounted on the front of the Enclosure are connected to the PCB male pin headers as follows:

- **Green LED:** Connected to the “RDY LED” pins on the PCB. The anode (longer lead) is connected to the + pin on the PCB and the cathode (shorter lead) is connected to the – pin on the PCB.

- Yellow LED: Connected to the + and – pins of the “ADMIT” pins on the PCB. The anode (longer lead) is connected to the + pin on the PCB and the cathode (shorter lead) is connected to the – pin on the PCB. The S and G pins on the PCB are not connected.
- Red Pushbutton: The + and – lugs on the back of the pushbutton are connected to the + and – pins respectively of the “REJECT” pin field on the PCB. The unmarked (slightly larger) lugs on the back of the pushbutton are the switch leads and are connected to the S and G pins of the “REJECT” pin field on the PCB. The order of these switch pins does not matter.

The IR Temperature sensor module is connected to the field marked “PN532” on the PCB, as follows:

- VIN on the sensor to +5v on the PCB.
- GND on the sensor to GND on the PCB.
- SCL on the sensor to SCL on the PCB.
- SDA on the sensor to SDA on the PCB.
- The IRQ pin on the PCB is not connected.

When the 3D printed parts are available and assembly is complete, the Top cover piece is mounted underneath a cabinet so that the assembled Stovetop Monitor electronics will be off to one side of the stovetop with the sensor aimed down at the stovetop from a 45 degree angle. After mounting the Top cover to the underside of the cabinet, the assembled Enclosure will affix under the Top by placing it up to and slightly to the rear of the Top. The Enclosure is then slid forward until it engages with the tabs of the Top. The Enclosure will now be held securely to the Top.

A M4 threaded brass insert should be heat-set into the hole at the rear of the top piece. A long M4 machine screw should be inserted into a threaded hole at the rear of the Top piece, as shown in figure 5, below. This “keeper” screw should be tightened by hand so that it holds the Enclosure body securely to the Top bracket, ensuring that the Enclosure can’t slide backwards and disengage from the Top.

After completing mounting the Stovetop Monitor Electronics, a USB cable should be inserted into the micro USB connector protruding from the rear of the Enclosure. The other end of the USB cable is connected either to a USB power supply or to a laptop computer in order to power the Electronics and, optionally, collect serial data from it. See the “Stovetop Monitor User Manual” for instructions for obtaining and using serial data to customize the Stovetop Monitor to a specific installation, stove type, and cooking routines.



Figure 5. Assembled Enclosure with Top Bracket and Keeper Screw.

STOVETOP MONITOR SOFTWARE.

The Stovetop Monitor software components are open source. The source files are in the “Software” folder in this repository:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Software

STOVETOP MONITOR FIRMWARE.

The Stovetop Monitor firmware is the software that is flashed to the Particle Photon 2 microcontroller that is part of the Stovetop Monitor Electronics. The source code is in this folder in this repository:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Software/Stovetop_Monitor

The main firmware is the “StovetopMonitor.ino” file. Note that the Adafruit library “Adafruit_MLX90614.h” is included and must be imported into the project in order to compile the “StovetopMonitor.ino” code.

In addition to the MLX90614 library, the following helper classes are included in “StovetopMonitor.ino” and must be included in the /src subfolder:

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- BtnStatusClass (.h and .cpp): helper class to monitor the pushbutton in a non-blocking and debounced manner.
- LEDcontrolClass (.h and .cpp): helper class to provide non-blocking control of LEDs.
- ElectricStovetopConstants.h: included file that provides the thresholds and timeouts for the installation that this project was developed for.
- TestStovetopConstants.h: this file can be included in lieu of “ElectricStovetopConstants.h” for firmware testing purposes. The thresholds and timeouts are lowered to allow the firmware to be tested without high temperatures and long timeouts.

The thresholds and timeouts that are used by the firmware have been included in a separate file so that they can easily be changed to meet the needs of different stovetops, installation details and cooking scenarios. The serial port monitoring capability of the firmware is provided so that a user can record cooking scenarios on their specific installation and then analyze the data captured in order to tune the thresholds and timeouts to their own unique situation. See the “Stovetop Monitor User Manual” for further information.

STOVETOP MONITOR WEBHOOK AND NOTIFICATION.

The Stovetop Monitor user can choose to receive Stovetop alarms on one or more mobile devices. The Photon 2 firmware publishes alarm events to the Particle Cloud whenever the firmware transitions into the ALARM state.

A Webhook must be created in the user’s Particle account that subscribes to these events. The Webhook is created using the Particle Console in the “Integrations” section. A JSON representation of a Particle Webhook that fires off a Pushover¹ notification on a user’s mobile device is included in this repository:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Software/Webhook/StoveTopAlarmWebhook.txt

The user must install the Pushover App on their mobile device. This App is available for both Android and iOS. Note that use of this App requires a one time in-app purchase of \$5 but does not require any recurring or ongoing fees.

The Webhook requires a user-specific “user access token” and a “pushover api token”. The “user access token” is created when a new user creates a Pushover account. The “pushover api token” is created on the Pushover website when the user creates an “app” on the Pushover cloud via the Pushover website. These tokens must be placed into the Particle Webhook where indicated by "<< your Pushover API token >>" and "<< your Pushover User access code >>".

¹ Pushover.net

Refer to the Particle and Pushover on-line documentation for further details.

STOVETOP MONITOR APP.

An Android App is provided with this project. The user may install this App on an Android device and subsequently use the App to remotely status their Stovetop Monitor. Figure 7, below, shows the main screen of this App.

NOTE 1: This App is based upon a template that allows the user to log into their Particle account, obtain a list of Particle devices in their account, and select the Particle Photon 2 device that is used on their Stovetop Monitor. The App Template was developed on an earlier project, see: https://github.com/TeamPracticalProjects/Particle_App_Template. Particle changed the way a user remotely logs into their account in June 2024 and the App only works with Particle accounts that were established prior to this date. An App that is based upon an updated template is not available at this time. Users with newer Particle accounts can utilize the Particle Console to remotely monitor their stovetop. See the “User Manual” for details.

NOTE 2: This App was developed in MIT App Inventor 2. In theory, the App could be built to run on iOS platforms as well. However, this requires an Apple Developer Account and publication of the App code to the Apple App Store. We have not presently chosen to do this and we cannot certify that the App can be built and work on iOS devices. iOS users can utilize the Particle Console to remotely monitor their stovetop. The Pushover alert mechanism is not affected by this. See the “User Manual” for details.

In order to install the App on an Android device, the .apk file must be downloaded to the user’s Android device. This file is located in:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Software/AI2_App

This folder also contains the MIT App Inventor 2 “source code” in the .aia file, in the spirit of open source.

After downloading “Stovetop_Monitor.apk” to the user’s Android device, the user must locate and tap on this file in order to install the App. Depending on the version of Android on the user’s device, some message about the App not being from a trusted source will appear. There will be an option to install the app anyway – choose this option to install the App on the Android device.

Tapping on the installed App will open the App and the screen will be similar to figure 7, below. However, the status indication at the bottom right of the screen will be red and indicate an error. This error occurs because the Photon 2 device in the user’s Stovetop Monitor must be selected before the App can be used to communicate with it. In order to select the device, tap the

“Setup” button at the lower left of the App home screen. This will take you to the setup screen, as shown in figure 6 below.

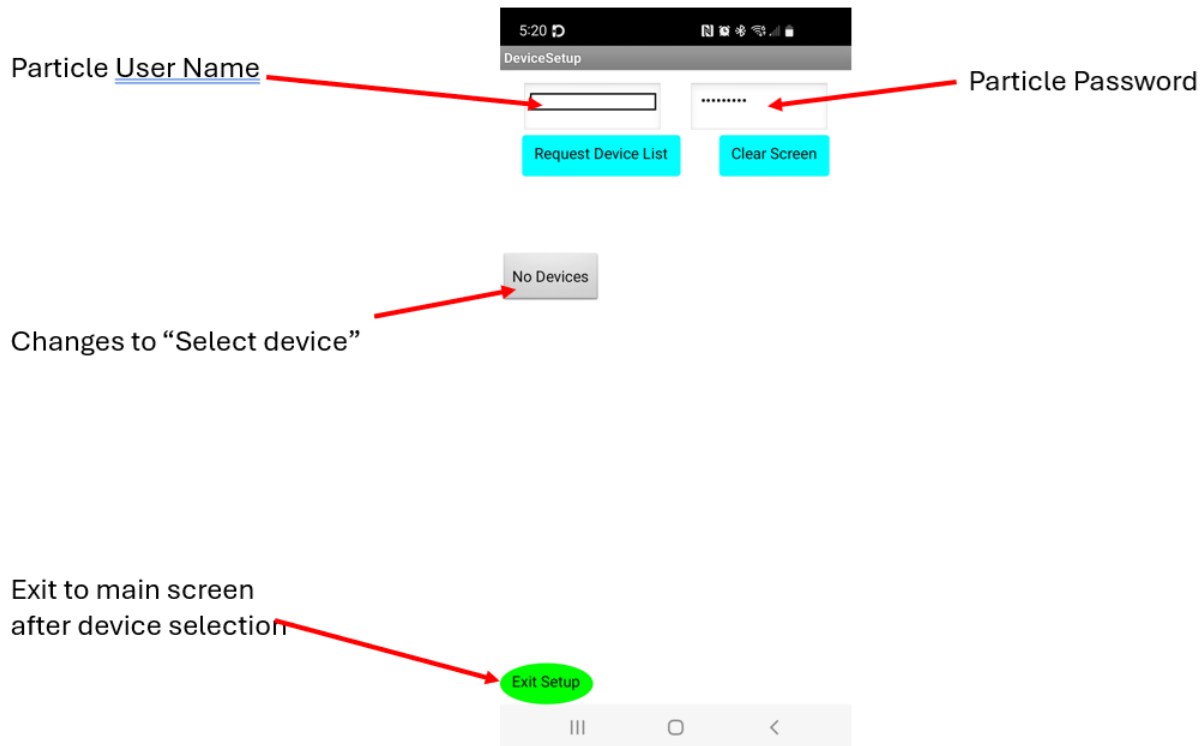


Figure 6. App Setup Screen.

Once in the setup screen of figure 6, type your Particle User Name and Particle Password into the text boxes indicated in figure 6. Then tap on “Request Device List”. If the login/password is correct, the button that says “No Devices” in figure 6 changes to “Select Device”. Tap on “Select Device” to view a popup list of all devices in the user’s Particle account. Tap on the Photon 2 device that is used on the Stovetop Monitor. Then tap on “Exit Setup” at the bottom left of the setup screen to return to the App home screen.

The App home screen should appear similar to figure 7, with the status at the bottom left being green and showing the selected Particle device to use with the App².

² Note that the Stovetop Monitor Particle Photon 2 device must be powered and on-line in order for the status to shown green; otherwise, the status will be red and indicate an error trying to connect to the device.

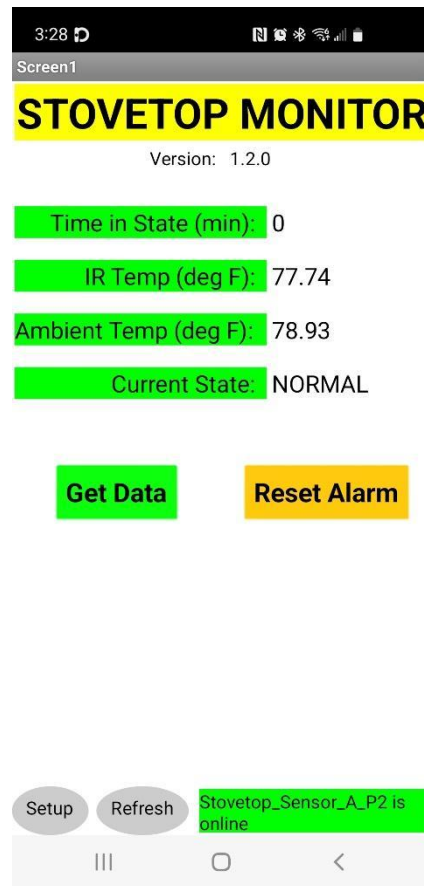


Figure 7. Stovetop Monitor App – After Device Selection.

See the “User Manual” for a description of using this App.